```
-----CMD-----
timidity Fanfare.mid --volume 200 -T 150
----/CMD-----
```

Ports and connections

When programs like virtual pianos and music creators do not produce sound, you have to change their connection by choosing the appropriate MIDI port. It has been observed that the Midi Through port doesn't play sound while Timidity ports work well. As an example, take the VMPK virtual piano. When it doesn't play sounds, you have to change its connections at *Edit→Connections*.

You can use connection programs like the Jack Audio Connection Kit to handle advanced audio connections including that of MIDI. They offer many complex facilities.

Sometimes the program shows that '…an instrument is not found.' This is not a problem of the program, but of the soundfont.

Soundfonts

We are familiar with text fonts. The text of a document is always the same, but the appearance depends on the fonts used while rendering (displaying). You might have noticed that the text of some websites appears differently in different browsers/devices. This is because they render those websites differently even though the content remains the same. As I mentioned earlier, MIDI files contain notations only. They might sound different in different platforms since the soundfonts used for rendering may be different.

Another problem with text fonts is that some regional language/special characters in documents cannot be displayed since the current font doesn't contain those characters. The same issue exists for soundfonts also. Some instruments are not available in some soundfonts, so a MIDI file containing parts for those instruments can't be played properly. Just as we solve the text font problem by using fonts with maximum languages/characters, we can use heavy soundfonts like Fluid, or combinations of different soundfonts to solve this MIDI problem.

By default, FreePats comes with TiMidity++ as the soundfont. Unfortunately, it is currently incomplete. So we have to install some other soundfont to enjoy MIDI files in clarity. Let us learn more about the installation and configuration.

Installation

Soundfonts come with the extension *.sf2*. They can be stored in any directory and used with suitable programs. But in order to use them with TiMidity++ (i.e., usual MIDI playback), we should have their configuration files in */etc/timidity*. Fortunately, we have got ready-to-install packages of some famous soundfonts. You may use a package manager to install them. The packages are *fluid-soundfont-gm, fluid-soundfont-gs* and *musescore-soundfont-gm*. The first two are very large, while the second one is lightweight.

Configuration

Even after installing a new soundfont, TiMidity++ uses the older one. We have to edit its configuration file to enable the new soundfont. Let us look at the Fluid GM soundfont as an example. First ensure that its configuration file is in */etc/timidity*.

For this, try the following command at the terminal:

```
ls /etc/timidity

fluidr3_gm.cfg  freepats.cfg  timidity_a.cfg
fluidr3_gs.cfg  timgm6mb.cfg  timidity.cfg
```

The output shows we have *fluid3_gm.cfg*, which is the configuration file of the Fluid GM soundfont. Now we have to add this into *timidity.cfg*, which is the configuration file of TiMidity++. Before doing that, keep a back-up copy of it in your home folder:

```
sudo cp /etc/timidity/timidity.cfg ~/timidity.cfg
```

Try the command given below to open the file:

```
sudo gedit /etc/timidity/timidity.cfg
```

> **Note:** You can use other text editors like *gvim* instead of *gedit*, depending on your distro.

We have to add this line at the end of the file:

```
source /etc/timidity/fluidr3_gm.cfg
```

But the file probably contains that line already, in a commented state. All you have to do is uncomment it (by removing the hash (#) in front of it). Now you may comment the line of the current soundfont by adding a # before it.

The configuration is complete and TiMidity++ starts to use the new soundfont.

> **Note:** You can use multiple soundfonts at a time by keeping their configuration lines uncommented. When both soundfonts contain the same instrument, the last soundfont is used. For advanced mixing of soundfonts, you have to edit their configuration files (e.g., */etc/timidity/fluidr3_gm.cfg)* or create a new soundfont using programs like Swami.

END

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